

What Can an Agent Know About Its Own Learning?

Endogenous Self-Modeling, Anchored Metacognition, and the Addressability
Bottleneck in a Developmental Self-Model Research Program

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Abstract

We ask whether an artificial agent can detect that its own learning is failing, estimate its own learning progress, and use that information to change its representational structure during its lifetime. We report the complete experimental record of a research program built around this question—including the parts that failed—and re-examine it under explicit evidence standards: every quantitative claim is traced to an output file or log, and each is labelled as demonstrated, suggestive, negative, hypothesis, or proposal. In homeostatic control tasks, an online self-model whose prediction error (“doubt”) is fed back to the controller detected 78% of severe hidden bodily faults within 15 steps (30% false alarms), and every evolved organism family that contained a self-model and doubt reached >90% healthy survival in all 6 of 6 evolutionary runs, versus 5 of 9 for comparison families (one-sided Fisher $p = 0.092$). This is suggestive, not conclusive, and detection did not by itself improve fault survival: a dynamic-programming ceiling shows the remaining gap is set by the reward, not by missing information. A re-execution of the original metacognitive agent shows that its previously reported perfect survival is uninformative (always resting also survives that setting), but that on harder settings its doubt-driven caution mode, not its hand-written rules, carries its survival (76.7 vs. 56.7% without doubt), at the price of mostly not working; when the same doubt is graded against the agent’s own predictions instead of measured outcomes, it stops detecting shocks (AUROC 0.46–0.51 vs. 0.87–0.92) and survival collapses (94.0 $\rightarrow$ 8.7% in the rule-free variant). In a developmental benchmark where task–cue–action mappings are re-drawn every lifetime, three successive dense local-plasticity learners failed to beat a cue-blind baseline. With a hand-specified sparse associative learner (no evolution), we find that adding *addressable* capacity matters and adding *units* does not: growth that imprints a new unit on each unrepresented pattern reached $58.6 \pm 0.6\%$ with 24 average units, matching a fixed 128-unit learner ($58.6 \pm 0.3\%$), whereas the same growth trigger with randomly wired units gave $50.5 \pm 0.0\%$ (fixed-16: $50.5 \pm 0.2\%$; cue-blind: 50.0%; 3 seeds $\times$ 400 paired lives). A few evolved runs (1–2 seeds per configuration) show the same ordering at a higher level (imprint growth 67.1%, fixed-128 64.5%, random growth 54.8%), which we treat as suggestive. However, the growth decisions in that result are driven by a novelty gate, not by metacognition; and a pre-specified prospective test shows that the program’s behavioural “learnability” signal tracks past progress ($r = 0.50$ – 0.58) but does not forecast future progress (partial r from -0.086 to -0.004 after controlling for recent accuracy; 6 runs). Our supported claim is therefore narrow: endogenous self-prediction error is useful, evolvable control information; representational addressability, not neuron count, is the bottleneck for developmental growth in this setting; and the agent’s estimate of its own learning progress is, so far, retrospective. We give a short theoretical account of why metacognitive signals must be anchored to externally generated outcomes, and a minimal program of pre-registered experiments that would be needed to support the stronger claim that metacognition can *cause* useful structural development.

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1 Introduction

Most learning systems separate a fixed architecture from the external optimizer that trains it. Biological learners do not: they keep models of their own changing bodies, modify synapses throughout life, allocate and retire representational resources, and appear to monitor whether their own learning is working [21, 61]. This paper reports a research program that tried to build artificial agents with some of these properties, and asks what the resulting evidence actually supports.

The guiding question is:

Can an artificial agent detect that its current learning process is failing, estimate its own learning progress, and use that information to modify or expand its own representational structure during its lifetime?

We decompose it into four progressively stronger questions:

- Q1** Can an agent learn a model of its own changing dynamics and derive a useful endogenous uncertainty signal from its errors?
- Q2** Can evolution produce local lifetime-learning rules rather than complete policies?
- Q3** Can an agent estimate whether its own behaviour is currently improving—and, more strictly, whether it *will* improve?
- Q4** Can persistent failure of learning trigger the construction of useful new computational structure?

What this paper is and is not. It is an evidence-first account of a sequence of experiments, several of which failed. It is *not* a demonstration of autonomous, metacognitively driven neurogenesis. Earlier drafts associated with this project reported results that we could not trace to code or outputs (Section 7); those numbers are withdrawn here and replaced with re-executed measurements. We think the resulting narrower paper is more useful than the original one: each failure removed a wrong explanation, and the remaining diagnosis is specific enough to be tested.

Contributions. Each is labelled with the strength of its evidence (Section 2).

1. **SUGGESTIVE Endogenous self-prediction error is useful and evolvable control information.** Surprise-based doubt detects most severe hidden faults; organisms with a self-model and doubt evolved to survive more reliably (6/6 vs. 5/9 runs, $p = 0.092$). Detection alone did not improve fault survival (Section 4).
2. **DEMONSTRATED An audit of the original metacognitive agent.** Its reported perfect survival is uninformative (always resting also survives). On hard difficulty, doubt-driven caution—not the hand-written rule layer—carries survival, by trading reward for safety (Section 4.3).
3. **DEMONSTRATED A negative result on dense local learners.** Three dense three-factor learners (V1–V3) did not beat a cue-blind baseline; their above-chance accuracy is explained by action priors (Section 6.3; V1–V3 numbers are reported from project logs and were not re-run).
4. **DEMONSTRATED Addressability, not unit count, limits growth** in a hand-specified sparse associative learner: imprint growth matches a fixed network $5\times$ larger on average, random-wired growth gives nothing, with paired lives and 3 seeds (Section 6.4). Evolution was not involved.
5. **NEGATIVE The learnability signal is retrospective.** A pre-specified prospective test shows it does not forecast future learning progress beyond recent accuracy (Section 6.5).

6. **SUGGESTIVE** **A theory of anchoring.** We formalize why self-evaluation must be tied to an externally generated reference (the Minimal Reference Condition), prove a sufficiency result for a linear case, and connect it to two concrete failures in this program (Section 5).
7. **PROPOSED** **A minimal decisive experimental program** with pre-specified hypotheses, controls, tests and falsification criteria (Section 9).

2 Evidence standards

Because the program produced results of very different strength, every substantive claim carries one tag:

Tag	Meaning
DEMONSTRATED	Measured in this paper or in committed result files, with the stated seeds and controls; the claim does not go beyond the measured setting.
SUGGESTIVE	A measured effect whose statistical support is weak (few seeds, confounds) or whose mechanism is not isolated.
NEGATIVE	A measured failure of a hypothesis or of a previously stated claim.
HYPOTHESIS	A claim we believe but have not tested.
PROPOSED	A design for an experiment that has not been run.
REPORTED	(Reported, not re-run.) A number reported in the project’s research log that we could not re-execute for this paper (the V1–V3 learners were run on a GPU; no per-run result files were committed). These numbers are used only descriptively.

Statistical conventions. Unless stated otherwise, “ $\pm$ ” is the standard deviation across seeds. Where the source tables report 95% confidence intervals from a t -distribution over 3 seeds, we say so; with $n = 3$ such intervals can exceed $[0, 100]\%$, which is itself informative about seed variance. Hand-built comparisons in Section 6.4 evaluate every configuration on the *same* sampled lives (paired design). We report p -values only with sample sizes and effect sizes. Appendix A maps every number in the abstract and in the claims section to its source file.

Materials. The program’s code lives in three branches of the project repository: the original agent (`main.py`); the learning-agent, hidden-fault, organism and ceiling experiments (branch `clau de/research-opinion-ly94jb`, commit `d185ccf`); and Benchmark B V1–V4 (branch `clau de/quirky-wright-6hfp0i`, commit `26cc897`), now maintained as the standalone repository `github.com/ismamsha/developmental-self-model`, which also holds the evolved V4 results and the research log (`docs/RESULTS.md`). All new analyses in this paper are in `paper/experiments/`.

3 Related and convergent work

Several of the program’s ideas were reached independently—often much earlier—by others. We list the closest matches explicitly (Table 1), both to credit prior work and because convergence is weak evidence that the ideas are natural.

What we have not found in prior work is the specific *combination* the program targets: an agent whose estimate of its own inability to learn triggers the allocation of new, addressable representations within one lifetime. We stress that this paper does not demonstrate that combination either.

Table 1: Ideas in this program and the closest independent work.

Idea in this program	Closest independent work
Internal (bodily) variables as first-class state; reward from homeostasis	Homeostatic RL [32]; homeostatic “feeling machines” [37]; interoceptive inference [51]; Ashby [6].
Learned self-model of one’s own body used for control	Resilient self-modelling robots [11, 34].
Prediction error as a control signal; doubt as precision	Active inference [22, 23]; dual control [7, 20].
Imagination of action consequences	Dyna [56], I2A [45], world models [28, 29]; ensembles for model uncertainty [12, 35].
Caution mode and emergency overrides	Shielding [3]; uncertainty-triggered abort/reset [17]; recovery policies [57]; constrained MDPs [1, 4].
Rules-only genomes: evolve plasticity rules, start from random weights	Evolved Hebbian rules in random networks [41]; evolved plastic networks [54]; decomposed plasticity under an information bottleneck [59]; genomic bottleneck [52, 65].
Three-factor local rules with eligibility traces	Gerstner et al. [25]; meta-learned three-factor rules [39].
Predicting one’s own learning progress	Learning progress as intrinsic motivation [27, 43, 49]; MAGELLAN [24].
Allocate a new unit when error is high <i>and</i> the input is novel	Resource-allocating networks [44] (the V4 imprint rule is essentially this, with a local three-factor readout); cascade correlation [18]; progressive / expandable networks [47, 63]; online growth and pruning [30]; integration of newborn units [36].
Utility-based pruning / replacement	Continual backpropagation [14].
Sparse expansion for pattern separation; fast associative memory	Babadi and Sompolinsky [8], Dasgupta et al. [13], Kanerva [31]; complementary learning systems [33, 40].
State that is itself a learner; nested timescales	TTT layers [2, 55], Titans [10], nested learning [9]; learning to learn in activations [15, 60].
Self-evaluation must be anchored (MRC)	Goodhart’s law [26, 38]; reward hacking and tampering [5, 16, 53]; self-modification and delusion boxes [42, 46]; Gödel machines [50]; Vingean reflection [19]; self-improving agents [62, 66]; self-rewarding LMs [64].

4 Part I — Endogenous self-modeling in homeostatic control

4.1 Environment

All Part I experiments use a homeostasis task (`HomeostasisEnv` in `main.py`). The agent has resources $r \in [0, 1]$ and degradation $d \in [0, 1]$, and chooses among REST (recover resources, efficiency $1 - d$), REPAIR (reduce d at a resource cost), and three WORK intensities (reward, at resource and degradation cost). Random shocks remove resources and add degradation. A life ends at 100 steps (survival bonus) or when $r \leq 0$ or $d \geq 1$ (death penalty). Difficulty presets (normal / hard / expert) set the shock probability (5 / 12 / 15%), shock size, recovery rate and costs. Seven of the ten observation dimensions are Gaussian noise.

4.2 The original agent (MSRA) and its claimed mechanism

The original Metacognitive Self-Referential Agent (MSRA) maintains a six-dimensional self-state $m_t = [x_t, r_t, c_t, g_t, d_t, \delta_t]$ (load, resources, capability, confidence, degradation, doubt), a learned self-model $\hat{m}_{t+1} = f_\theta(o_t, m_t, a_t)$, and doubt as a leaky integral of the self-prediction error,

$$\delta_{t+1} = \min(\delta_t + 0.15, 0.7 \delta_t + 0.3 \|m_{t+1} - \hat{m}_{t+1}\|_1/6), \quad (1)$$

with decay and failure-driven boosts. Doubt raises an “anxiety” gain β and, above a threshold, switches the agent into a caution mode that masks risky actions. Actions are chosen by imagining the next self-state under each action and scoring it with a composite utility.

4.3 Audit of the original agent

Reading `main.py` alongside the preprint that described it revealed several discrepancies (listed in Section 7). Three matter for interpretation:

1. `select_action` contains a hand-written rule layer that runs *before* any metacognition: never work when $r < 0.6$ (rest, or repair if $d > 0.75$); emergency rest/repair on falling resources; forced low-intensity work after long rest streaks. Imagination is further filtered by hard-coded “expert economics” that encode the environment’s true costs.
2. The actor-critic network is trained but never selects actions; its training update does not detach the advantage and can diverge to NaN.
3. `main.py` imports a module (`helpers`) that was never committed; we supply no-op stand-ins (as the research branch also does).

We therefore re-ran the agent with ablations that separate its components (`msra_experiments.py`): the full agent; the same with doubt clamped to zero (*no doubt*); without the rule layer (*no shield*); without the rule layer and without the hard-coded economics (*learned only*), with and without doubt; the rule layer alone with no learning in the loop (*shield only*); the base imagination agent; the actor-critic network acting on its own; random and always-rest policies; and two *self-graded* variants in which the self-model is scored and trained against its own predictions for the externally measured variables (used in Section 5). Each runs on all three difficulties, 3 seeds, 100 online training episodes.

Findings. All numbers are from Table 2 (last 50 of 100 episodes, 3 seeds).

1. **NEGATIVE Normal difficulty does not discriminate.** Every variant that rests when resources are low survives 100%, including always-rest. A 100% survival rate in this setting says nothing about metacognition. On expert difficulty nothing survives reliably (full agent $5.3 \pm 4.2\%$; always-rest $9.3 \pm 4.2\%$). The preprint’s comparative claims therefore have no regime in which they reproduce.
2. **DEMONSTRATED The rule layer is not what keeps the agent alive.** On hard difficulty, removing it leaves survival unchanged (76.0 ± 8.7 vs. $76.7 \pm 2.3\%$), and the rules alone survive $51.3 \pm 12.9\%$, no better than always resting ($54.7 \pm 2.3\%$). This corrects the claim in an earlier version of this draft that the agent’s survival came from its rules.
3. **DEMONSTRATED Doubt-driven caution carries survival, by not working.** Clamping doubt to zero lowers hard survival from 76.7 to $56.7 \pm 6.4\%$ in the full agent, and from 94.0 ± 2.0 to $8.0 \pm 3.5\%$ in the variant without rules and hard-coded economics. The mechanism is blunt: the

Table 2: Re-execution of the original agent (`main.py`) with ablations. Survival (%) over the last 50 of 100 online training episodes, mean $\pm$ SD over 3 seeds. Reward, caution (% of steps in caution mode), work (% of work actions) and AUROC are for hard difficulty; AUROC measures how well the prediction error the agent *uses* separates shock steps from other steps (0.5 = no information). Source: `paper/results/msra_audit.json`.

Variant	Normal	Hard	Expert	Reward	Caution	Work %	AUROC
Full agent	100.0	76.7 $\pm$ 2.3	5.3 $\pm$ 4.2	21.9 $\pm$ 2.2	91.4 $\pm$ 2.0	19 $\pm$ 1	0.90
doubt clamped to 0	100.0	56.7 $\pm$ 6.4	6.0 $\pm$ 3.5	41.6 $\pm$ 1.2	–	34 $\pm$ 1	0.81
without rule layer	100.0	76.0 $\pm$ 8.7	10.7 $\pm$ 4.2	24.4 $\pm$ 2.8	83.1 $\pm$ 3.3	17 $\pm$ 2	0.89
Rule layer alone	100.0	51.3 $\pm$ 12.9	1.3 $\pm$ 2.3	33.0 $\pm$ 2.2	–	33 $\pm$ 2	0.84
No rules, no coded economics	100.0	94.0 $\pm$ 2.0	8.7 $\pm$ 4.2	7.3 $\pm$ 0.5	53.0 $\pm$ 3.2	3 $\pm$ 0	0.90
doubt clamped to 0	99.3 $\pm$ 1.2	8.0 $\pm$ 3.5	0.0	1.4 $\pm$ 1.9	–	26 $\pm$ 2	0.82
Base imagination agent	100.0	83.3 $\pm$ 11.0	8.7 $\pm$ 3.1	7.1 $\pm$ 0.8	–	3 $\pm$ 0	0.83
Actor-critic acting	19.3 $\pm$ 7.0	0.0	0.0	3.0 $\pm$ 0.5	–	58 $\pm$ 1	–
Full, self-graded doubt	100.0	48.7 $\pm$ 11.0	3.3 $\pm$ 1.2	36.9 $\pm$ 2.5	0.7 $\pm$ 0.5	33 $\pm$ 3	0.51
No rules, self-graded doubt	36.0 $\pm$ 17.4	8.7 $\pm$ 9.9	0.7 $\pm$ 1.2	9.2 $\pm$ 9.9	16.3 $\pm$ 9.1	43 $\pm$ 20	0.46
Random policy	18.0 $\pm$ 5.3	0.0	0.0	3.1 $\pm$ 0.3	–	58 $\pm$ 1	–
Always rest	100.0	54.7 $\pm$ 2.3	9.3 $\pm$ 4.2	0.7 $\pm$ 0.2	–	0	–

agent is in caution mode on 53–91% of steps and works on 3–19% of them, so reward falls (full agent: 41.6 $\rightarrow$ 21.9). The most cautious variant survives best and earns least (94.0%, reward 7.3). This is a survival–reward trade-off, not a free improvement; the learned MSRA-L agent (Table 3) reaches 99.5% on hard with reward 68.5.

4. **DEMONSTRATED** The thresholds make caution nearly permanent. With thresholds of 0.02 (enter) and 0.005 (exit) on a mean absolute error over six dimensions, caution is active on 70% of steps even on normal difficulty; `main.py`’s own training loop gives 91.6% (normal) and 98.2% (hard, 80% survival over 30 episodes), versus 4.4% reported in the preprint.
5. **NEGATIVE** The actor-critic network, acting on its own, diverges (thousands of NaN policy evaluations per run) and performs like the random policy.

The self-graded rows are discussed in Section 5.3.

4.4 A learned agent (MSRA-L)

Because the original agent’s behaviour is largely hand-coded, a learned variant was built in which every decision comes from learned components [58]: a double DQN over internal state; an ensemble of five probabilistic self-models predicting next state, reward and probability of death; doubt as ensemble disagreement; β as a multiplier on the predicted probability of death (“fear of death”); optional Dyna-style imagination. Evaluation: 400 training episodes, then 200 greedy episodes, 3 seeds; “ $\pm$ ” in Table 3 is the 95% t -interval over seeds as reported in the result files.

Findings. **SUGGESTIVE** A learned probability of death, amplified by β , is what produced reliable survival on hard difficulty (78.5% DQN $\rightarrow$ 99.5–100%), while also raising reward. **NEGATIVE** Ensemble-disagreement doubt added little: the dynamics are stationary and fully observed, so the self-model quickly becomes certain. **NEGATIVE** Imagination hurt under large, rare shocks (expert: 1.5% with, 33.7% without), plausibly because each Gaussian self-model smears rare large shocks into frequent small ones, so imagined experience understates risk. Expert results vary across seeds from 5 to 49%; no expert-mode conclusion is supported.

Table 3: MSRA-L survival (%) after training (200 greedy evaluation episodes, 3 seeds, 95% t -interval). Source: `results/summary.md`, branch `research-opinion-ly94jb`. On normal difficulty every policy, including always-rest, survives 100%.

Policy	Hard: survival	Hard: reward	Expert: survival	Expert: reward
Always rest	57.2	1.1	8.3	2.6
Four-line threshold rule	53.5	33.4	3.7	18.1
DQN (value learning only)	78.5 ± 10.2	65.6 ± 12.9	28.8 ± 44.2	19.1 ± 4.7
MSRA-L without doubt	73.0 ± 37.5	60.3 ± 6.7	0.5 ± 0.0	14.9 ± 5.4
MSRA-L with doubt	81.0 ± 23.7	58.9 ± 4.9	1.0 ± 3.3	16.8 ± 10.1
MSRA-L doubt + fear of death	99.5 ± 1.2	68.5 ± 1.4	1.5 ± 1.2	19.3 ± 4.1
Doubt + fear, no imagination	100.0 ± 0.0	73.0 ± 4.2	33.7 ± 60.6	24.9 ± 8.3

Table 4: Severe hidden faults (500 training episodes, 300 greedy evaluation episodes, 3 seeds, 95% t -interval). Detection: the doubt signal exceeds its healthy 95th percentile within 15 steps of fault onset; false alarm: same test at a random step of a healthy episode. Source: `results/faults_severe/summary.md`.

Agent	Healthy	Fault	Fragile fault	Detected ≤ 15 steps	False alarms
Always rest	60.7	27.2	—	—	—
Surprise-doubt (metacognitive)	99.6 ± 0.9	43.0 ± 55.8	27.2 ± 23.2	78.4 ± 16.3	29.6 ± 9.8
Oracle (told the fault)	100.0 ± 0.0	52.7 ± 36.7	27.8 ± 14.0	—	—
Fault-blind	99.5 ± 2.0	71.7 ± 6.7	45.4 ± 19.2	5.2 ± 19.3^a	1.6 ± 5.5^a

^a Using ensemble disagreement as the signal.

4.5 Hidden faults: detection is not use

To create a situation in which self-knowledge should matter, a hidden fault strikes in half of all episodes at a random step in [20, 70] (hard difficulty). *Severe* faults: a motor fault (work drains an extra $0.5\times$ intensity) or a fragile body (shocks $3\times$ as frequent). Agents share the same learned machinery and differ in one extra input: *surprise-doubt* (a running average of signed prediction error in units of predicted spread), the true fault flag (*oracle*), or nothing (*fault-blind*).

Findings. **DEMONSTRATED** Surprise-doubt is informative: it detects 78% of severe faults within 15 steps at a 30% false-alarm rate, while ensemble disagreement barely reacts (5%). Ensemble disagreement measures uncertainty *about* the model; surprise measures the model being *wrong about the world*—only the latter changes when the body changes. **NEGATIVE** Detection did not improve survival. Even the oracle survived faults less often than the fault-blind agent (52.7 vs. 71.7%). A plausible explanation—not isolated experimentally—is that the extra input splits 500 episodes of experience into a rare “broken” regime that the value network learns poorly, while the fault-blind agent learns one robust policy and reacts to the resource levels it can see. All agents reduce work by a similar amount after a fault (Figure 1, right). With mild faults, all learned agents were within noise of each other (fault survival 94.6–95.1%).

4.6 Evolved organisms and rules-only genomes

The metacognitive loop was then moved inside one small network trained by evolution only (OpenAI-ES [48]; 400 generations, population 128, 32 lives per genome per generation; test: 600 new lives). Plastic weights are $W + A \odot H$, with H updated during life by a three-factor rule (pre $\times$ post $\times$ neuromodulator); a readout predicts resource and damage changes and learns during life by the delta rule; doubt is a running average of its error and is fed back as input; a neuromodulator

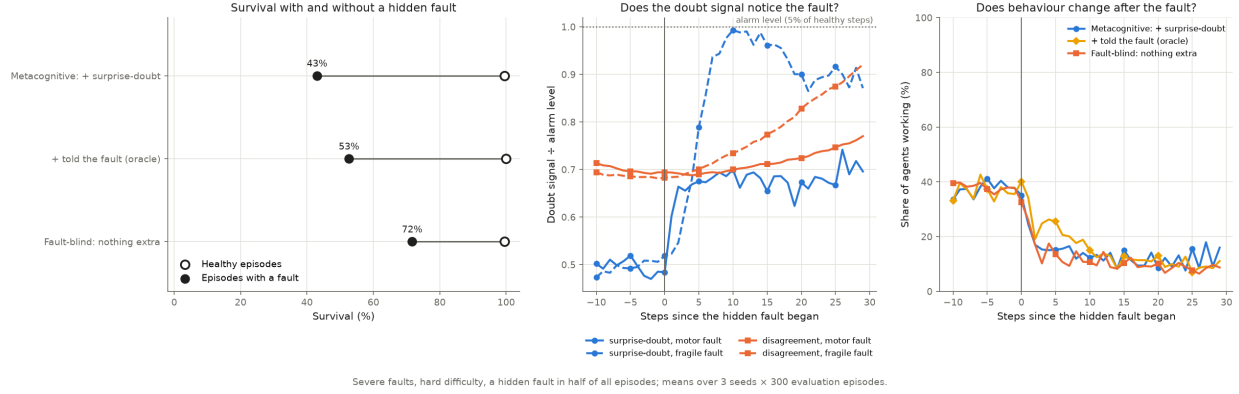


Figure 1: Severe hidden faults (from the research branch). Left: survival. Middle: surprise-doubt rises sharply after onset of a fragile fault and moderately after a motor fault; ensemble disagreement drifts slowly. Right: all agents reduce work similarly after a fault, whether or not they detect it.

Table 5: Evolved organisms, severe faults, hard difficulty. Per-seed healthy / fault survival (%) on 600 test lives. A run “succeeds” if healthy survival exceeds 90%. Source: `results/organism_severe/*.json`.

Organism	Genome	Seed 0	Seed 1	Seed 2	Successes
Innate wiring	288	14.2 / 5.9	0.0 / 0.0	99.0 / 72.1	1/3
Innate + told the fault	320	8.8 / 7.9	99.3 / 74.7	97.2 / 73.4	2/3
Innate + self-model and doubt	736	99.7 / 68.4	100.0 / 73.0	99.7 / 73.4	3/3
Plastic wiring, no doubt	583	8.4 / 0.0	100.0 / 65.8	97.6 / 73.7	2/3
Full organism (doubt gates plasticity)	1095	99.3 / 67.4	100.0 / 71.7	100.0 / 68.6	3/3
Rules only (random wiring at birth)	161	67.6 / 0.0	85.1 / 0.7	97.6 / 48.1	1/3

unit reads doubt and gates plasticity. The *rules-only* organism inherits no weights: each life starts from random wiring and an empty self-model, and the genome holds only per-neuron learning-rule coefficients (161 numbers).

Findings. SUGGESTIVE All six runs of the two self-model-and-doubt families succeeded, versus 5 of 9 runs of the three comparison families with inherited or plastic wiring (one-sided Fisher exact test, $p = 0.092$; we recomputed this). Three caveats bound this claim. (i) With three seeds per family, the test is underpowered. (ii) Genome size is confounded: the doubt families have larger genomes (736 and 1095 vs. 288–583 parameters). (iii) The *successful* comparison runs reach the same level as the doubt families (e.g. innate seed 2: 99.0 / 72.1; plastic seed 2: 97.6 / 73.7). The effect, if real, is on the *reliability* of evolution, not on the attainable level of performance. NEGATIVE In the full organism the evolved neuromodulator barely responds to faults (Figure 2, right): evolution used doubt as an input for action selection, not as a signal to rewire. SUGGESTIVE The rules-only organism can work: one of three runs reached 97.6% healthy survival from random wiring, using only inherited learning rules. The same run survived only 48.1% of faulty lives, and the other two runs did not succeed. This is a proof of possibility, consistent with Najjarro and Risi [41], not evidence of robustness.

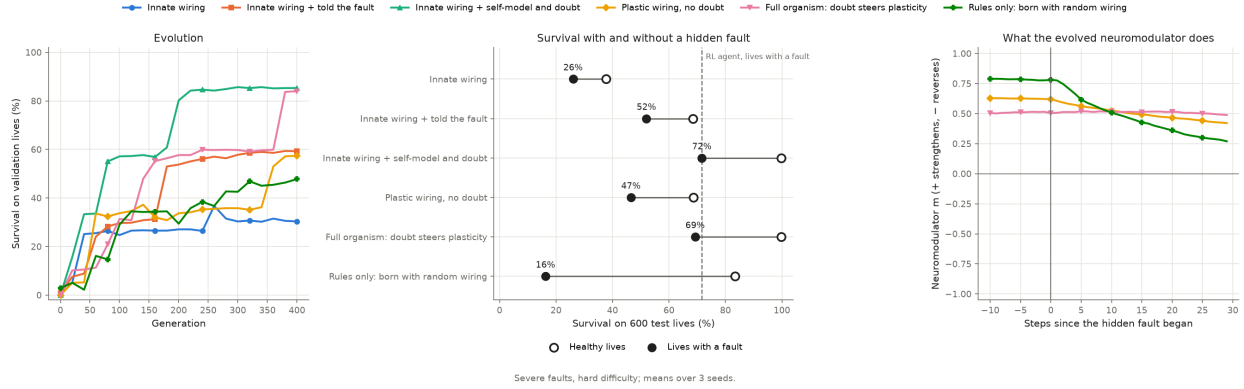


Figure 2: Evolved organisms (research branch). Left: validation survival during evolution (seed means). Middle: healthy vs. faulty test lives. Right: the evolved neuromodulator after fault onset—roughly flat in the full organism.

4.7 The benchmark ceiling and the role of reward

A dynamic-programming analysis over the known body dynamics (`survival_ceiling.py`, re-run for this paper) gives the best survival any policy could reach once a severe fault strikes, even with instant knowledge of the fault and full health at onset: 100.0% for the motor fault, 72.4% for the fragile fault, 86.2% overall. **DEMONSTRATED** The best agents already reach $\approx 100\%$ on the motor fault; all headroom is in the fragile fault, where the best agents reach 39–47% (organisms) and 45% (fault-blind RL). Surviving a fragile fault requires stopping work entirely, and the reward makes continued work worth the risk. Two lessons follow and shaped everything after: a benchmark ceiling must be computed before an intelligence gain is claimed, and self-knowledge cannot rescue an agent from an objective that rewards the behaviour that kills it.

5 Part II — Anchoring: why self-evaluation needs an external reference

Two failures in this program have the same structure. In Benchmark B V2, “learnability” was defined from changes in the self-model’s own prediction error, and it could rise while behavioural accuracy fell (**REPORTED**; Section 6.3). And the earlier write-ups of the original agent evaluated it with statistics it partly generated itself. A companion note proposed a general principle, the Minimal Reference Condition (MRC). We restate it, give a proof for a linear case (the original note gave none), and test it.

5.1 Setting

Definition 1 (Self-modifying evaluator). *A system has parameters θ_t and an internal evaluator E_t . It updates θ to increase E_t and may also update E_t . True performance T is fixed and not directly observed. The reference gap is any measure of divergence between E_t and T ; in the linear case $E_t(\theta) = w_t^\top \theta$, $T(\theta) = v^\top \theta$, it is the angle $\Delta_t = \angle(w_t, v)$.*

Proposition 1 (No guarantee without a reference). *If the evaluator update may select any E_{t+1} from a class containing, for every value c , a function equal to c at θ_{t+1} regardless of $T(\theta_{t+1})$, then no rule that accepts updates by the criterion $E_{t+1}(\theta_{t+1}) > E_t(\theta_t)$ can guarantee $T(\theta_{t+1}) \geq T(\theta_t)$.*

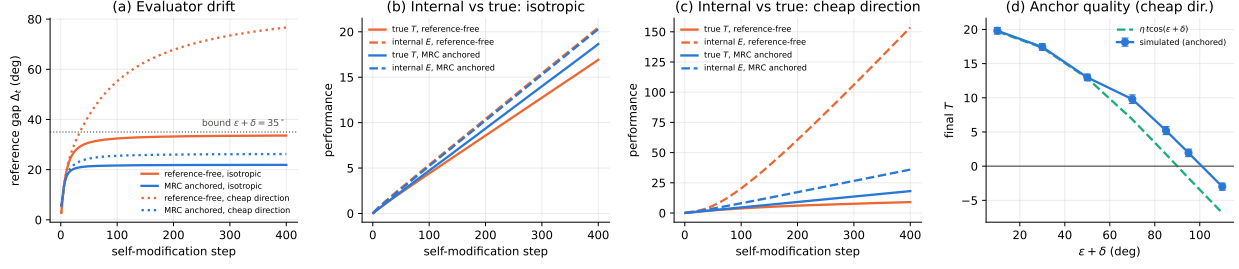


Figure 3: Linear model of evaluator drift (200 seeds). (a) Reference gap. (b, c) Internal score E (dashed) vs. true performance T (solid). (d) Final T under MRC as the anchor loosens, against the lower bound of Proposition 2.

Proof. Take any θ' with $T(\theta') < T(\theta_t)$ and choose E' from the class with $E'(\theta') = E_t(\theta_t) + 1$. The pair (θ', E') satisfies the acceptance criterion and decreases T . $\square$

Remark. The proposition is close to a tautology; its value is in naming the assumption that must be broken. It is an instance of Goodhart’s law [26, 38] and of reward tampering [16], specialized to an agent that can rewrite its own evaluator.

Definition 2 (Minimal Reference Condition, linear form). *A fixed unit vector R with $\angle(R, v) \leq \delta$ exists and every evaluator is projected onto the cone $C(R, \epsilon) = \{w : \angle(w, R) \leq \epsilon\}$ after each update.*

Proposition 2 (Sufficiency of MRC for expected improvement, linear case). *Let $\theta_{t+1} = \theta_t + \eta P \hat{w}_t + \xi_t$ with $\hat{w}_t = w_t / \|w_t\|$, $\eta > 0$, zero-mean noise ξ_t , and a symmetric positive semi-definite preconditioner P with $Pv = v$. Under MRC with $\epsilon + \delta < \pi/2$,*

$$\mathbb{E}[T(\theta_{t+1}) - T(\theta_t) \mid \theta_t, w_t] \geq \eta \cos(\epsilon + \delta) > 0, \quad \text{so} \quad \mathbb{E}[T(\theta_t)] \geq T(\theta_0) + \eta t \cos(\epsilon + \delta).$$

Proof. $\mathbb{E}[T(\theta_{t+1}) - T(\theta_t)] = \eta v^\top P \hat{w}_t = \eta (Pv)^\top \hat{w}_t = \eta v^\top \hat{w}_t = \eta \cos \angle(w_t, v)$. By the triangle inequality for angles, $\angle(w_t, v) \leq \angle(w_t, R) + \angle(R, v) \leq \epsilon + \delta < \pi/2$, and cosine is decreasing on $[0, \pi]$. Summing over t gives the second statement. $\square$

The condition $Pv = v$ allows the policy to move much more easily in directions orthogonal to v —cheap, “cosmetic” directions—without breaking the guarantee, because the anchored evaluator never rewards them enough to dominate.

5.2 Simulation

We simulate the linear model ($n = 20, 400$ steps, $\eta = 0.05$, noise $\sigma = 0.1$, 200 seeds; `mrc_toy.py`). The evaluator self-endorses: $w \leftarrow w + 0.05 \hat{\theta}$. We run an isotropic policy ($P = I$) and a policy with a cheap direction $h \perp v$ ($P = I + 9hh^\top$).

Findings. DEMONSTRATED (for the model; this is a toy, not evidence about real systems.) Without a cheap direction, the reference-free evaluator drifts to a gap of $33.6^\circ \pm 5.4^\circ$ but true performance still rises (final $T = 16.9$ vs. 18.7 anchored): drift alone does not produce collapse, correcting a claim in the original note. With a cheap direction, the reference-free gap grows to $76.7^\circ \pm 5.3^\circ$ and the internal score inflates to 153.6 while true performance reaches only 9.0 ; the anchored system keeps the gap at 26.2° (max 31.7° , below the 35° bound), with $T = 18.1$. Across anchor quality, final T stays above the bound of Proposition 2 (Figure 3d).

5.3 Anchoring in a metacognitive agent

HYPOTHESIS A metacognitive signal is useful only if part of what it is scored against is generated outside the agent’s own modelling machinery. Surprise-doubt compares predictions to measured bodily change, which the agent cannot rewrite; V2’s self-model-based learnability did not, and drifted; V3’s behavioural learnability is anchored to task outcomes.

DEMONSTRATED The self-graded variants of the original agent test the simplest case (Table 2). In them, the self-model’s error for the externally measured variables (load, resources, degradation) is computed—and the self-model trained—against its own prediction instead of the measured value; control still sees the true state. The only change is therefore what the evaluator is anchored to. On hard difficulty, the prediction error the agent uses stops carrying information about shocks (AUROC 0.51 and 0.46, versus 0.90 for both anchored counterparts), caution almost never triggers in the full agent (0.7% of steps), and survival falls from 76.7 to $48.7 \pm 11.0\%$ (full agent) and from 94.0 to $8.7 \pm 9.9\%$ (rule-free variant). This is the reference gap of Section 5 in its most direct form: a self-evaluation that is free to agree with itself agrees with itself. Two caveats: the manipulation also changes the self-model’s training targets, so the collapse combines an uninformative doubt signal with a self-model that never learns the measured dynamics; and the original agent’s “actual” state vector is itself half self-generated (c_t, g_t, δ_t are the agent’s own variables), so even its anchored doubt mixes an anchored and an unanchored component.

6 Part III — Developmental Benchmark B

6.1 Why a new benchmark

The homeostasis task gives an agent no reason to grow: a 32-unit controller is enough, so a rational developmental system should learn not to grow, and the absence of growth there falsifies nothing. Benchmark B was designed so that capacity could matter:

- each life draws new random mappings $(q, c) \mapsto a^*$ from task q and cue c to action, so the genome cannot memorize them and must encode *how to learn* them (default 4 tasks $\times$ 4 cues $\times$ 4 actions; chance 25%);
- trials are delayed: task and cue are visible at an instruction step and hidden at the response step;
- tasks accumulate over four phases, with the newest task dominant and a rehearsal fraction of older tasks;
- hidden body faults can occur, and there is an energy/damage economy;
- the organism starts small and may grow or prune units; lifetime learning is local (no back-propagation); evolution (ES) tunes the genome.

6.2 Benchmark integrity audit

Accuracy above 25% is not evidence of associative learning. We audited the benchmark code for shortcuts; Table 6 lists what we found.

6.3 V1–V3: dense local learners fail to learn the mapping

All numbers in this subsection are **REPORTED** (research log `docs/RESULTS.md` and the project draft; GPU runs; no committed per-run result files).

Table 6: Shortcut and integrity audit of Benchmark B (V3/V4 code).

Risk	Present?	Consequence and remedy
Class imbalance within a phase	Yes. Four cues map to four actions independently, so collisions are common; the best constant action per phase scores $\approx 50\%$.	Any learner can exceed chance by learning a phase action prior. Report the cue-blind margin, not accuracy vs. chance.
Time / phase observable	Yes. Normalized time $t/(T - 1)$ is an input, so phase identity is decodable.	Phase-conditioned priors are learnable. The hindsight per-phase cue-blind baseline covers this; an online cue-blind learner with the time input should also be run.
Task-frequency prior	Yes. The newest task gets 70% of trials (V4 meta test: 50%).	Favours recency; report per-task retention (task0 columns).
Reward leakage	Previous trial’s perceived reward is visible at the next instruction.	Enables win-stay/lose-shift across trials; cues are re-drawn every trial, so this helps only as a prior. Keep, but ablate.
Action-dependent cost	Yes. Energy cost rises with the action index.	Survival pressure favours low-index actions independently of the cue; fitness can conflict with accuracy. Report accuracy separately from fitness.
Fault leakage	Faults act through visible energy/damage.	Legitimate: detecting a fault from its bodily effects is the point.
Hindsight baseline	The cue-blind phase baseline chooses the best action per phase <i>in hindsight</i> .	It is an upper bound on cue-blind strategies with phase knowledge, not an online agent; margins above it are conservative.
Degenerate control	In V4 the no-plasticity control is bit-identical to random policy (zero readout $\rightarrow$ uniform policy, same random stream).	It does not test birth wiring in V4; V3’s no-plasticity control (25.3%) does.
Delay length	One recurrent step between instruction and response.	A short delay: memory is barely tested.

V1. Random birth wiring, recurrent state, local plastic weights, self-model, doubt, a learnability estimate, and HOLD/GROW/PRUNE decisions with reserve neurons. With six actions (chance 16.7%), accuracy stayed at 16–17% from generation 0 to 50 while the phenotype underwent ≈ 35 structural changes per life and drifted toward ≈ 10 neurons. **NEGATIVE** Structural change without a working learner is noise. Diagnosis: no reliable lifetime learner; lethal cognitive errors; structural decisions far too frequent; delayed reward not credited to the instruction-time state.

V2. Four cues and actions (chance 25%), eligibility traces across the delay, reward-gated three-factor plasticity, stochastic softmax responses, a rehearsal curriculum, non-lethal cognitive errors, a slower structural clock, and a genome seeded near a reward-modulated Hebbian rule. Final evaluation: 35.5% accuracy (phases 39.7 / 35.2 / 34.1 / 33.1%), 99.5% survival, a within-life rise from $\approx 28\%$ to $\approx 40\%$ followed by decline as tasks accumulated, and essentially no growth (16.0 average neurons; 0.26 structural changes per life). **NEGATIVE** No developmental growth.

V3. Per-trial eligibility reset (V2’s traces credited $\approx 72\%$ of each reward to earlier trials with different cues), reward-prediction-error neuromodulation, an $\text{onehot}(a) - \pi$ output rule, evolvable temperature, recurrent gain normalization, stronger cue input, behavioural learnability and capacity pressure. Controls: random policy 25.06%; no plasticity 25.34%. Fixed-16 evolution (100 generations): 37.7% at generation 0, 39.3% final (survival 99.7%). Cue-blind lifetime baseline 39.9%; cue-blind per-phase baseline 49.7%; perfect-learner ceiling $\approx 82\%$. The within-life trace rose during phase 1 ($30.3 \rightarrow 38.5 \rightarrow 44.2 \rightarrow 51.7\%$) and dropped to 39.8% at the start of phase 2. **NEGATIVE** 39.3% < 49.7%: the V3 learner did not demonstrate $(q, c) \mapsto a$ learning; its behaviour is consistent with learning a phase-dependent action prior that is overwritten when the task distribution changes.

V3 diagnostics. **REPORTED** (hand-built rule, 200 lives; docs/RESULTS.md). A per-life ridge probe on response-time states decoded the *cue* at 93.9% (baseline 24%) and the $\text{task} \times \text{cue}$ conjunction at 63.4% (baseline 8.9%), while the organism scored 37.6% against a 49.5% cue-blind baseline. Removing the delay (cue visible at response) raised accuracy only to 40.6–43.7% (probe $\approx 98\%$ / $\approx 71\%$); leaky units lowered it to 34.3%; a centered readout gave 40.2%. Interpretation: retention across the one-step delay is not the main bottleneck; the information is largely present, but dense reward-modulated updates of shared weights cannot bind 16 conjunctions from ≈ 7 exposures each without mutual overwriting. These diagnostics are single-configuration runs without seed statistics.

Learnability in V3. Behavioural learnability (a fast-minus-slow accuracy trend predicted from the organism’s features) was reported to rise while accuracy improved ($\approx +0.42$ at 51.7% accuracy) and fall below zero (≈ -0.10) after task transitions, while capacity pressure rose. We test the prospective version of this claim in Section 6.5.

6.4 V4: a sparse associative learner and functional growth

The V3 diagnosis—information present, dense readout interfering—led to V4 (`dsm_benchmark_b_v4.py`): the instruction’s task+cue input is projected through random birth wiring to a pool of 128 potential units, of which the active ones compete in a k -winners-take-all code ($k = 1$); the code is held across the delay and read out by a fast associative matrix trained with the same local three-factor rule, $\Delta W = \eta m_t z (\text{onehot}(a) - \pi)^\top$, with m_t driven by reward-prediction error (and optionally doubt and learnability). *Imprint growth* recruits a dormant unit whose input weights are set to the current pattern, when the controller decides to grow and no active unit already owns that pattern (novelty gate); *random growth* recruits a dormant unit with its random birth wiring. The configurations *full*, *nolearn* and *basic* use imprint growth plus pruning; *nolearn* removes learnability and *basic* removes all self-model signals from the controller, gate and modulator.

All V4 results below use a *hand-specified* genome—no evolution—and were re-run for this paper on CPU: 3 seeds $\times$ 400 lives (standard task) or 200 lives (metacognition test), every configuration on the same lives. Standard task: 4 tasks $\times$ 4 cues $\times$ 4 actions, 240 steps, faults in 65% of lives.

Findings.

1. **DEMONSTRATED Capacity matters in this learner.** Accuracy rises monotonically with fixed size ($46.3 \rightarrow 58.6\%$ from 8 to 128 units; seed SD ≤ 0.3). This is the precondition that made growth worth studying, and it did not hold for the dense learners.
2. **DEMONSTRATED Addressability, not unit count.** Random-wired growth uses on average 18 more units than imprint growth yet gives no improvement over fixed-16 (50.5 vs. 50.5%), and

Table 7: V4, standard task, hand-built genome (mean $\pm$ SD over 3 seeds; each seed 400 paired lives). Cue-blind = best constant action per phase in hindsight (50.0%); perfect-learner ceiling 80.4% (seed 0). Source: [paper/results/v4_standard/](#).

Configuration	Accuracy	Margin	Task 0, ph. 4	Avg. units	Final units	Grows
Random policy	24.9 $\pm$ 0.1	-25.1	24.8	16	16	0
Fixed-8	46.3 $\pm$ 0.2	-3.7	67.0	8	8	0
Fixed-16	50.5 $\pm$ 0.2	+0.5	74.4	16	16	0
Fixed-32	53.8 $\pm$ 0.1	+3.8	77.7	32	32	0
Fixed-64	56.7 $\pm$ 0.3	+6.7	83.4	64	64	0
Fixed-128	58.6 $\pm$ 0.3	+8.6	86.0	128	128	0
Random-wired growth from 16	50.5 $\pm$ 0.0	+0.5	56.0	41.8	67.8	51.8
Imprint growth from 16	58.6 $\pm$ 0.6	+8.6	87.5	23.9	29.8	13.8
Full (imprint + prune)	58.4 $\pm$ 0.6	+8.4	87.4	23.9	29.6	13.9
Full without learnability	58.6 $\pm$ 0.5	+8.6	87.5	24.0	29.7	14.0
Full without self-model signals	56.4 $\pm$ 0.4	+6.4	83.5	23.4	29.0	13.2

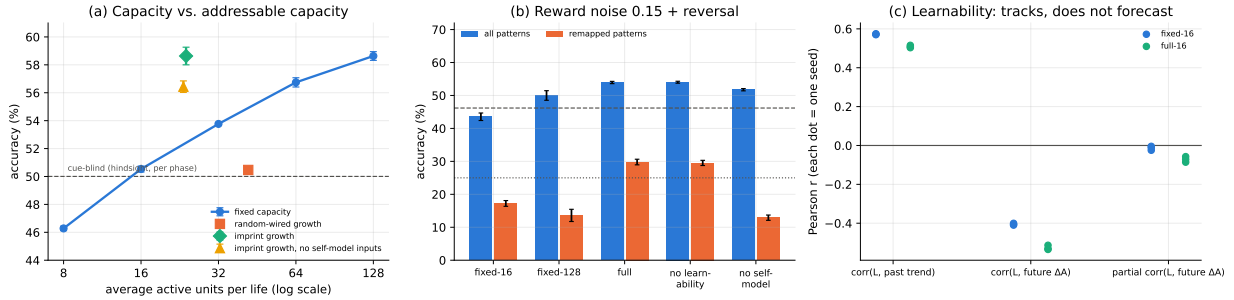


Figure 4: V4 (hand-built genome, 3 seeds). (a) Accuracy vs. average active units: fixed capacity (line), random-wired growth, imprint growth, and imprint growth without self-model inputs. Dashed: hindsight cue-blind baseline. (b) Reward noise 0.15 plus mid-life remapping of two tasks: overall accuracy and accuracy on remapped patterns. Dashed: cue-blind; dotted: chance. (c) Learnability validation (Section 6.5).

worse retention of task 0 (56.0 vs. 74.4%). Imprint growth reaches the fixed-128 level (58.6 vs. 58.6%) with 23.9 average units, and the best task-0 retention (87.5%). The per-seed differences imprint – random are +8.0, +8.9 and +7.7 points on identical lives.

- NEGATIVE** **Metacognition is not what drives growth here.** Removing learnability changes nothing (58.6 vs. 58.4%); removing all self-model signals costs 2.0 points. GROW decisions occur at slightly *lower* capacity pressure than HOLD decisions after an error (difference -0.032 to -0.036 across seeds): the novelty gate, not pressure, decides. In the standard task every error signals “this pattern has no slot yet”, so novelty is a sufficient trigger.
- NEGATIVE** **Even fixed-16 in V4 does not clear the cue-blind baseline** (+0.5 points). Capacity-limited sparse learning is still close to the shortcut regime; only larger or growing networks clear it by a margin (≈ 8.5 points), which is still far below the 80.4% ceiling.

The imprint rule is, in essence, a resource-allocating network [44] with a local three-factor readout: its success confirms an old idea in a new setting rather than establishing a new mechanism.

Evolved genomes. **SUGGESTIVE** The repository also contains V4 runs in which ES tuned the genome (population 64, 32 lives per genome, 100 generations; the 53-parameter genome without

Table 8: V4, evolved genomes (ES, 100 generations), standard task. One row per committed run. Cue-blind 49.7–49.8%; survival 99.7–100% in every run; random-growth units are the mean of the two seeds. Source: `results/v4_standard/*_seed*.json` in the developmental-self-model repository.

Configuration (seeds)	Accuracy	Avg / final units	Phase 1 → 4 (seed 0)	Task 0, ph. 4
Fixed-16 (0, 1)	54.2 / 54.2	16 / 16	63.9 → 46.4	75.2 / 73.3
Fixed-32 (0, 1)	58.7 / 58.6	32 / 32	67.1 → 51.8	79.5 / 79.3
Fixed-64 (0, 1)	62.5 / 61.6	64 / 64	69.6 → 56.7	84.0 / 83.6
Fixed-128 (0, 1)	64.8 / 64.2	128 / 128	70.9 → 60.4	86.4 / 86.0
Random growth (0, 1)	55.0 / 54.6	38.7 / 62.8	62.2 → 49.8	57.7 / 59.5
Imprint growth (0)	67.1	23.9 / 29.6	62.7 → 69.1	85.7
Full (0)	67.0	23.9 / 29.6	62.6 → 68.7	85.6
Full without learnability (0)	66.8	23.9 / 29.5	62.5 → 68.6	84.9
Full without self-model signals (0)	67.1	23.7 / 29.4	62.9 → 68.9	85.4

Table 9: V4.1, reward noise 0.15 + reversal, hand-built genome (3 seeds × 200 paired lives). Cue-blind 46.2%; ceiling 90.0%. Source: `paper/results/v4_meta/`.

Configuration	Accuracy	Remapped	Task 0, ph. 4	Avg. units	Repairs/life
Random policy	25.2 ± 0.2	25.1	23.8	16	0
Fixed-16	43.5 ± 1.1	17.2 ± 0.9	15.3	16	0
Fixed-128	50.0 ± 1.5	13.6 ± 1.8	11.8	128	0
Full-16	53.9 ± 0.3	29.8 ± 0.9	36.9	24.8	18.8
Full-16 without learnability	54.0 ± 0.3	29.5 ± 0.8	35.9	24.8	18.7
Full-16 without self-model signals	51.7 ± 0.3	12.9 ± 0.8	8.8	24.4	0.3

repair and gate genes). Table 8 lists every committed run. Evolution raised every configuration (fixed-16 from 50.5 to 54.2%, now 4.5 points above cue-blind), and the ordering of Table 7 is preserved: imprint growth (67.1%, 23.9 average units) exceeds fixed-128 (64.5%), random growth stays near fixed-16 (54.8%), and imprint accuracy *rises* across phases (62.7 → 69.1%) where fixed networks decline (fixed-128: 70.4 → 60.6%). With one seed for every growth configuration, these are suggestive only: seed-to-seed spread in the two-seed rows is ≤0.9 points, but a single evolutionary run can differ from another in ways the evaluation spread does not capture. Evolved *full*, *nolearn* and *basic* are indistinguishable (67.0, 66.8, 67.1%), confirming that self-model signals are redundant in the standard task.

Metacognition under ambiguous errors. To create conditions in which an error is *not* always a missing slot, V4.1 adds reward noise (15% of feedback flipped; fitness scores the true outcome) and a mid-life remapping of two tasks, plus a REPAIR action that resets the slot owning the current pattern, with a hand-set connection from a per-slot surprise signal (“slot doubt”: recent confident-error rate of that unit) to REPAIR and to a plasticity gate (480 steps, rehearsal 0.5, no faults).

DEMONSTRATED (hand-wired) A per-slot surprise signal lets the organism find and reset slots whose mapping changed: accuracy on remapped patterns is 29.8% with it and 12.9% without it (paired lives, 3 seeds; per-seed differences +17.3, +16.3, +17.1). Because the ablation removes the signal from REPAIR, from the gate and from the modulator together, the effect cannot yet be assigned to one lever; and because the REPAIR connection was set by hand, this shows that the signal is *usable*, not that evolution would discover its use. **NEGATIVE** Again learnability contributes nothing (54.0 vs. 53.9%). Note also that all learners except the repairing ones score *below chance*

Table 10: Prospective validation of learnability (seeds 0 / 1 / 2; 95% life-level bootstrap intervals in the source file `paper/results/learnability_validation.json`).

Run	$r(L, \text{trend})$ [S1]	$r(L, \Delta A)$ [S2]	partial r [S3]	$r(\text{tr.}, \Delta A)$
Fixed-16	0.575 / 0.570 / 0.572	-0.400 / -0.410 / -0.406	-0.011 / -0.025 / -0.004	-0.30
Full-16	0.515 / 0.508 / 0.503	-0.513 / -0.531 / -0.535	-0.057 / -0.072 / -0.086	-0.35

on remapped patterns: they keep executing the old mapping. **NEGATIVE** The benefit depends on the condition. In a smaller per-condition check (**REPORTED**; 40 lives $\times$ 3 seeds, docs/RESULTS.md) full – basic was +1.9 (standard), -2.0 (noise only), +5.2 (reversal only) and +2.6 points (noise + reversal): under noise alone, the hand-set repair policy fires on noise and hurts. Whether evolution tunes this away is untested.

6.5 Is the learnability signal a forecast? A pre-specified prospective test

The strongest version of Q3 is that the agent predicts its *future* learning progress. The V3/V4 learnability signal L_t is trained online to predict $\tanh(10(\bar{a}_t^{\text{fast}} - \bar{a}_t^{\text{slow}}))$, a *retrospective* trend of recent accuracy. We therefore tested it prospectively (`learnability_validation.py`; criteria written in the script header before it was run). For each life and response step t with $W = 10$: $\Delta A_t = \text{mean}(\text{correct}_{t+1..t+W}) - \text{mean}(\text{correct}_{t-W+1..t})$. Criteria: **S1** $\text{corr}(L_t, \text{trend}_t) > 0.3$ (tracks the past); **S2** $\text{corr}(L_t, \Delta A_t) > 0$ with a life-level bootstrap 95% CI excluding 0 (forecasts); **S3** the same for the partial correlation after regressing both on recent trend, $1 - \bar{a}^{\text{slow}}$, past-window accuracy and phase (adds information). All three had to hold in every seed. V4 hand-built genome, fixed-16 and full-16, 3 seeds $\times$ 400 lives ($\approx 40,400$ samples per run). In fixed-16, learnability is a passive readout that does not affect behaviour, so the test has no feedback loop.

NEGATIVE S1 passes; S2 and S3 fail in every run. The signal tracks past progress, and it correlates *negatively* with future progress—a recent upswing predicts regression to the mean. After controlling for trivial statistics its partial correlation with future progress is zero (fixed-16; all three bootstrap intervals include 0) or slightly negative (full-16; intervals exclude 0 on the negative side). An event study reproduces the qualitative V3 pattern— L_t is highest just before a task transition (0.24 in fixed-16) and decays to ≈ 0 within 14 responses—but the decay lags the accuracy drop, as a retrospective signal would. **Consequence:** the claim “the agent represents that its behaviour is improving” is supported only in the retrospective sense. This is not surprising given the training target; a forecasting version would have to be trained on a delayed target (the realized ΔA once $t + W$ is reached), which is local and causal (Section 9, E8).

6.6 Why naive GROW-vs-HOLD comparisons mislead

The V3/V4 code compares accuracy on the next occurrences of the same pattern after GROW vs. HOLD decisions. In the V4 standard runs this gives 36–38% after GROW and 53–55% after HOLD (all seeds), which read naively says growth *hurts*. It does not: GROW is chosen on first encounters with a pattern (novelty gate), HOLD mostly on patterns that already have a slot. The comparison is dominated by selection; the between-condition comparison in Table 7 (same lives, growth enabled or not) is the valid causal estimate. A decision-level causal estimate requires randomizing the structural action (E9).

7 Reviewer-2 audit of this program’s earlier claims

Table 11 lists statements from earlier drafts and write-ups that do not survive scrutiny, and what replaces them here. We include it because the project’s credibility depends on it.

Table 11: Audit of earlier claims.

Earlier claim	Problem	Status here
MSRA: 100% survival vs. 67% (MBRL), 73% (PPO), 81% (fixed β), 86% (PPO-Lagrangian); ablation and sensitivity tables; $t(29) = 15.32$, Cohen’s $d = 2.41$	No baseline, ablation or statistics code or output exists in any branch.	Withdrawn. Replaced by the re-executed audit (Table 2).
MSRA’s 100% survival demonstrates the value of metacognition	On normal difficulty always-rest also survives 100%; on expert the agent survives 5.3%.	Uninformative; the discriminating regime (hard) shows doubt helps survival at a large reward cost.
MSRA switches to caution “without explicit hard-coding”	<code>select_action</code> applies hand-written thresholds before metacognition; imagination is filtered by hard-coded environment costs.	False; corrected (Section 4.3).
Doubt: $\gamma = 0.95$, thresholds 0.15/0.05; 5% shocks	Code: 0.7/0.3 mixing, thresholds 0.02/0.005; the run configuration uses expert difficulty (15% shocks).	Replaced by the code’s values (Eq. 1).
Doubt- β correlation $R^2 = 0.89$	β is defined as a function of doubt; the correlation is true by construction.	Removed.
Phase portrait, shock-response and long-horizon figures	Not produced by any committed code.	Removed.
Parameter counts: self-model 12,422, actor-critic 8,961	Actual: 11,462 and 14,790.	Corrected.
MSRA spends 4.4% of steps in caution mode (figure and preprint)	<code>main.py</code> ’s own training loop gives 91.6% (normal) and 98.2% (hard) with the committed code and no-op stand-ins for the missing module.	Not reproduced; reported as such.
Actor-critic “policy”	Trained but never selects actions; update can diverge to NaN.	Reported (Table 2).
MRC: “we prove” the anchored gap stays bounded; “feedback loop causes w_t to rotate away”	No proof was given; in the isotropic model drift saturates and T still rises.	Proof given for the linear case (Prop. 2); collapse shown to require a cheap direction.
Self-model + doubt organisms “evolved more consistently”	$n = 3$ per family, $p = 0.092$; genome-size confound; successful comparison runs reach the same level.	Kept as SUGGESTIVE with all three caveats.
Rules-only reached $\approx 98\%$	One of three seeds, 97.6% healthy, 48.1% with faults.	Kept as proof of possibility with the full per-seed record.

Earlier claim	Problem	Status here
V3 learnability shows the organism knows “my behaviour is improving”	Tested only retrospectively.	Prospective test fails (Section 6.5).
Capacity pressure drives growth	In V4, GROW occurs at lower pressure than HOLD.	Growth is novelty-driven in the tested setting.
V2 accuracy above chance indicates learning of the task	Cue-blind baseline not computed.	Reinterpreted via V3 controls as action-prior learning.
No-plasticity control validates birth wiring (V4)	Bit-identical to the random-policy control in V4.	Valid only for V3.

8 Claims

Level A — supported now. *In small synthetic benchmarks, (i) an online self-model’s prediction error is informative about hidden bodily change and is associated with more reliable evolution of successful controllers (suggestive; 6/6 vs. 5/9 runs, $p = 0.092$), although detection by itself did not improve survival; (ii) for lifetime structural growth, what matters is whether new capacity is addressable: with a hand-specified sparse associative learner, imprinting new units on unrepresented patterns matched a fixed network five times larger in average size, while equally triggered growth of randomly wired units gave no benefit (3 seeds, paired lives; single evolved runs show the same ordering at a higher level); (iii) the growth decisions in that result are driven by novelty, not by metacognitive estimates; a per-unit surprise signal becomes useful for repairing remapped associations when errors are ambiguous; (iv) the agent’s behavioural learning-progress signal tracks past progress but does not forecast future progress; and (v) a doubt signal is useful only while it is anchored to measured outcomes: graded against the agent’s own predictions, it loses its information about disturbances and the agent’s survival collapses.*

Level B — within reach of one or two experiments. *An organism can forecast its own near-future learning progress from local signals (E8), and structural allocation triggered by that forecast (persistent error $\times$ low predicted progress $\times$ novelty or interference) improves learning over matched HOLD and novelty-only controls under conditions where novelty alone is insufficient (E9–E10), with the controller’s use of these signals discovered by evolution rather than set by hand.*

Level C — the long-term DSM hypothesis. HYPOTHESIS *An artificial learner can discover during its lifetime that its present way of representing an experience prevents it from learning it, and respond by constructing the computational structure needed to learn it—with the learning rules, the self-model and the structural policy all inherited as a compact genome.*

Verdict. **Suitable now as a workshop, negative-results or diagnostic paper; requires a small number of decisive experiments for a main-track claim.** The threshold is concrete: a main-track paper on metacognitively driven development would need (a) a learnability signal that passes S2 and S3 of Section 6.5 in ≥ 5 seeds, and (b) a randomized structural intervention showing that growth triggered by that signal beats novelty-only growth and matched HOLD in a condition where novelty is not sufficient, with the controller evolved rather than hand-set. Neither exists yet. What exists—the audit, the negative results, the addressability control and the anchoring analysis—is defensible now.

9 A minimal decisive experimental program

PROPOSED All experiments use Benchmark B (V4 code unless stated), paired lives across conditions, ≥ 5 seeds (more where the effect is small), and a fixed evaluation set of 1,000 lives per seed. Success criteria are stated before running. The general statistical plan: paired differences per seed; mean, SD and all per-seed values; Cohen’s d_z on paired differences; two-sided paired t -test (or Wilcoxon signed-rank when $n < 8$ makes normality untestable) with Holm correction within each experiment; for per-step analyses, bootstrap over lives. Stopping: each experiment runs its pre-set budget; no early stopping on results.

Table 12: Decisive experiments. IV: independent variable; DV: dependent variable.

Exp.	Hypothesis	IV / DV / controls	Support / falsify	Failure modes
E1 Representation probe (V3 dense) (<i>partly done, 1 config.</i>)	Task $\times$ cue is linearly decodable at response time.	IV: size 16/32/64, delay on/off. DV: ridge-probe accuracy per life (held-out trials), measurement only. Controls: probe on shuffled labels; probe on birth-state network. 5 seeds.	High probe + low accuracy $\Rightarrow$ readout/credit bottleneck (reported: cue 93.9%, task $\times$ cue 63.4%). Low probe even without delay $\Rightarrow$ representation bottleneck.	Probe overfits (use held-out trials); decodable but not linearly separable by a local rule.
E2 No-delay (<i>partly done</i>)	Recurrent retention is not the main bottleneck.	IV: delay vs. cue visible at response. DV: accuracy, cue-blind margin. 5 seeds.	Large gain without delay $\Rightarrow$ retention bottleneck; small gain supports interference diagnosis.	Cue visibility adds input drive, not only memory; match input norms.
E3 Leaky retention (<i>partly done</i>)	A local leaky unit rescues delayed performance if retention matters.	IV: leak $r \in \{0, 0.5, 0.8\}$. DV: accuracy, probe accuracy. 5 seeds.	Improvement proportional to E2 gap.	Leak increases cross-trial interference.
E4 Oracle learner (GRU, backprop)	The benchmark is learnable at 16–64 units with the available exposure.	(a) Meta-trained GRU (RL ² style [15]) over many lives, learning within life in activations; (b) tabular ideal learner. DV: accuracy vs. 80.4% ceiling, cue-blind margin. 5 seeds.	GRU at $\geq 60\%$ with 32 units $\Rightarrow$ capacity/exposure are adequate and local learners are the bottleneck. GRU near cue-blind $\Rightarrow$ redesign benchmark.	Meta-training budget too small; report training curves to saturation.
E5 Fast associative memory (no growth)	Sparse code + local readout beats cue-blind at fixed size.	Done hand-built (Table 7) and evolved with 2 seeds (Table 8). Remaining: ≥ 5 seeds.	Evolved fixed-32 $\geq$ cue-blind +5 points in all seeds.	Evolution degrades a good hand rule (compare to generation 0).
E6 Cue-blind baselines	Claimed mapping learners beat the strongest cue-blind learner.	Add an <i>online</i> cue-blind learner that sees time and task but not cue; keep hindsight baseline.	Margin > 0 over both, per seed.	Online baseline weaker than hindsight; report both.
E7 Capacity scaling	Performance increases with fixed capacity.	Done for hand-built V4 (8–128 monotone). Repeat with evolved genomes and the V3 dense learner.	Monotone increase in ≥ 4 of 5 seeds.	Flat curve $\Rightarrow$ capacity is not the target; growth has nothing to fix.

Exp.	Hypothesis	IV / DV / controls	Support / falsify	Failure modes
E8 Learn-ability forecast	A locally trained forecaster predicts future ΔA .	IV: target = retrospective trend (current) vs. delayed realized ΔA (new). DV: S1–S3 of Section 6.5; calibration curve of L vs. realized ΔA ; within-phase analysis. 5 seeds.	S2 and S3 pass in all seeds. Falsified if the delayed-target forecaster also has partial $r \leq 0$.	Predictability of ΔA is itself near zero (compute the ceiling with an offline regressor on the same features).
E9 Growth causal test	At matched eligible states, GROW improves subsequent learning and retention vs. HOLD.	Randomize the structural action at eligible decisions (error + novel pattern) with $p = 0.5$; DV: accuracy on the next 3 occurrences of the pattern, old-task retention, learning speed. 5 seeds $\times$ 1,000 lives.	Positive average treatment effect with CI excluding 0; supports causal usefulness of growth.	Randomization changes capacity trajectory; also report whole-life effects.
E10 Growth ablations	Metacognitive triggers beat novelty-only triggers when novelty is insufficient.	Configs: fixed-small, fixed-large, random growth, raw-error trigger, novelty trigger, pressure trigger (with E8 forecaster), full. Conditions: standard; noise + reversal; interference-inducing overlapping cues. Evolved controllers, 5 seeds.	Full > novelty-only in the ambiguous conditions (paired, Holm-corrected); equal in standard. Falsified if novelty-only matches full everywhere.	Evolution does not use the signals (inspect controller weights); triggers encode phase/time (ablate time input).

Falsification of the program. The DSM direction should be revised or abandoned if: an evolved associative learner cannot beat the online cue-blind baseline; a meta-trained GRU also cannot; fixed capacity from 16 to 64 does not help under evolved rules; a delayed-target learnability forecaster fails S3; randomized growth has no positive effect; growth helps only by effectively creating a large network immediately; growth decisions can be predicted from phase or time alone; or a fixed memory system without metacognition achieves the same with less compute.

10 Limitations

- All evidence comes from small synthetic tasks; nothing here implies competence on language, robotics or open-ended problems.
- Most comparisons use three seeds. Paired designs and small seed-to-seed variance make the V4 contrasts clear, but the organism and fault results remain statistically weak.
- V1–V3 numbers are reported from logs; we could not re-execute them. The V4 results with seed statistics use a hand-built genome; evolved V4 results exist for only 1–2 seeds per configuration.

- Growth in V3/V4 activates units from a pre-allocated pool; it reduces nothing in compute.
- Terms such as genome, doubt, anxiety, neurogenesis and neuromodulator are computational analogies, not biological claims.
- The anchoring theory is proven only for a linear model with a fixed true objective.
- The `main.py` audit uses 100 online episodes per run (the original used 30 for its figure and 200 in its default configuration).

11 Conclusion

We set out to show that an agent could notice that its own learning was failing and grow the structure it needed. The evidence supports three narrower statements. Self-prediction error is useful and evolvable information about one’s own body, though knowing is not the same as acting on it. When a lifetime learner grows, new capacity helps only if it is addressable—imprinted on what the learner currently cannot represent—and in our setting a simple novelty gate suffices to decide when. And the agent’s estimate of its own learning progress is, for now, a summary of the past rather than a forecast. Self-evaluation must be anchored: the same doubt signal that keeps an agent alive when it is scored against measured outcomes becomes uninformative, and the agent fragile, when it is scored against the agent’s own predictions. The stronger claim, that an agent’s forecast of its own inability to learn can *cause* useful structural development, remains an open hypothesis with a concrete, small set of experiments that would test it.

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Reproducibility

The paper, its analysis scripts and their outputs are in the `paper/` directory of `github.com/ismamsha/developmental-self-model`. New analyses: `paper/experiments/` (`learnability_validation.py`, `mrc_toy.py`, `fig_v4.py`, `summarize_msra.py`, and `msra_experiments.py`, which needs the original `main.py`; set `MSRA_REPO` to a checkout of `github.com/ismamsha/Self-Referential-AI-with-Proactive-Imagination-Metacognition`). Outputs and the exact commands: `paper/results/README.md`. The V4 code is `benchmarks/dsm_benchmark_b_v4.py`, identical to commit 26cc897 of the MSRA repository apart from paths in its docstring. Evolved V4 runs and the research log are in the same repository. Part I results (MSRA-L, hidden faults, organisms, ceiling): branch `claude/research-opinion-ly94jb` of the MSRA repository.

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A Evidence ledger

Number	Source
78% detection, 30% false alarms; fault survival 43.0 / 52.7 / 71.7% 6/6 vs. 5/9 successful runs; per-seed survival; genome sizes Fisher $p = 0.092$	<code>results/faults_severe/summary.md</code> (research branch) <code>results/organism_severe/*_seed*.json</code> , field final recomputed: <code>scipy.stats.fisher_exact([[6,0],[5,4]], 'greater')</code>
Ceiling 100.0 / 72.4 / 86.2% MSRA-L table <code>main.py</code> audit	<code>survival_ceiling.py -faults severe</code> , re-run <code>results/summary.md</code> (research branch) <code>paper/results/msra_audit.json</code> ; per-run records in <code>paper/results/msra_audit_raw.tar.gz</code>
Parameter counts 11,462 / 14,790 V1–V3 numbers	counted from <code>main.py</code> modules project research log (draft of September 2026); not re-run
V4 standard table	<code>paper/results/v4_standard/hand_comparison.json</code> , <code>v4_standard.txt</code>
V4 evolved table	<code>results/v4_standard/*_seed*.json</code> (developmental-self-model repository)
V3 diagnostics; V4.1 per-condition gaps	<code>docs/RESULTS.md</code> (developmental-self-model repository); not re-run
<code>main.py</code> own training loop (91.6 / 98.2% caution) V4.1 table	<code>paper/results/main_py_crosscheck.txt</code> <code>paper/results/v4_meta/hand_comparison.json</code> , <code>v4_meta.txt</code>
GROW vs. HOLD pressure and next-occurrence accuracy	<code>growth_analysis</code> fields in the V4 JSONs
Learnability S1–S3, event study	<code>paper/results/learnability_validation.json</code> , <code>.txt</code>
Anchoring simulation	<code>paper/results/mrc_toy.json</code>